Co-Design of a Computer Aided Design (CAD) Object Evaluation Interface for Course Evaluators

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Computer aided design is a core fundamental skill in the engineering curriculum. In the current world, "Industry 4.0" dominates the way design and manufacturing goals are taught to engineering students. A revamp in the curriculum also demands a revamp in the evaluation processes. Thus, there exists a need to overcome traditional challenges that CAD evaluators constantly face – enormous time investments, repetition, and lack of standardization. This paper attempts to address that by presenting a novel web-based method for evaluating CAD parts, developed by co-designing with CAD evaluators (Graduate Teaching Assistants, and Professors primarily). The tool is then evaluated by evaluators and tested for overall complexity in use, intuitiveness, and time consumption.

CCS CONCEPTS • **Hypertext languages** • **Web-based interaction** • **User interface programming**

Additional Keywords and Phrases: Computer-Aided Design, CAD, Evaluation, Design Intent

1 INTRODUCTION

Computer Aided Design has been a core curriculum item in engineering education. In the modern world, the focus on engineering education lies with training students to be industry ready and meet the goals aligned with the concept of "Industry 4.0". [1] This brings together an effort to redesign the curriculum for engineering education to focus on developing the "digital thread". [2] The "digital thread" essentially is an exact digital replica of a manufactured part, with all the relationships and contexts established within a digital ecosystem, where the entire life cycle of the part is recorded, tracked, updated, and maintained. This enables organizations to quickly iterate for improvements and testing, without needing to invest a lot in associated costs.

The major gap in getting learners to be ready for Industry 4.0 goals directly affects the teachings in an engineering classroom. Traditionally, engineering was an apprenticeship heavy skill, where learning was done during the job. [3] As time progressed, the shift from apprenticeship to formal education methodologies (such as traditional classroom instruction) has had a significant impact on the way engineering is taught. The current approach is conceptual and tool heavy, moving away from a real-world industry scenario, which focuses on designing for reusability [4] and interdisciplinary project-based tasks. [5] Thus, with a new generation of learners, designing for reuse and mimicking industry processes, the standard evaluation methodologies fall short, since they focus on whether the end product is correct or not, rather than the design intent. "Design Intent" can be defined as the function of the part, i.e. how should the final part representations should behave and evolve over its lifetime, rather than just look correct. [27] Design intent is the core driver of design for reuse, and with the current system of manual evaluation, there are three challenges that need to be tackled: the large amount of time invested to evaluate parts manually (i.e. going through each feature one at a time) [26], having to refer to multiple requirements documents per part i.e. the lack of standardization [21, 25], and the repetitive nature of the rubric yet each part being unique. [24]

The paper attempts at a solution for the provided problem statement, specifically for course evaluators. In this paper, academic evaluators comprise of Professors, Graduate Teaching Assistants, and any individual with grading or evaluating duties. The primary research question and discussion points are:

- Is there a significant difference in the overall time taken to evaluate a CAD model, between the manual method (typing scores and feedback) and the evaluator co-designed interface?
  The author's hypothesis is that there is less time taken to evaluate when using the co-designed interface.
- Discuss the design implications for building such an evaluation tool that facilitates knowledge transfer with new evaluator onboarding

The research question and discussion directly attempt at addressing the challenges faced by current evaluators (in an academic setting), in a large-scale course that follows the standards set by Industry 4.0 goals. In such a course, the evaluators comprise a background of interdisciplinary engineering fields, thus mimicking the need to create up-to-par tools for evaluators as well (along with the learners). The goal and contribution of the paper is to produce industry ready students, who can themselves end up as evaluators, thus standardizing the way CAD information is interpreted and evaluated within a multi-disciplinary context. [5,6]

2 RELATED WORK

The modern era of engineering education is centered around the "Industry 4.0" concept related to the fourth industrial revolution, that focuses on digitization of the manufacturing ecosystem, with computer aided techniques. [1] The field of engineering is regarded as a highly technical field, focusing on developing the analytical, conceptual, and results-focused nature of learning. [7] However, modern-day engineering is focused on additional aspects of interdisciplinary learning, problem solving, decision making, and inculcating a consistent urge to learn, which facilitates the core goals behind "Industry 4.0". [1,8]

2.1 HCI in Computer Aided Design (CAD)

The field of "Product Design" is driven by Computer Aided Design. [16] These software are often accompanied by the conventional implementation of the WIMP (Windows, Icons, Menus, Pointer) interface, that relies heavily on the user's behavioral patterns and mental model of the application. [17] However, the mental model and the spatial model may not always align, thus needing the user to translate 3D space information to a 2D movement scheme. [18]

Niu et. al. [17] in their paper research the concept of natural HCI for CAD. The concept, first introduced by Steve Mann, [19] involves using natural human behavior (speech, gestures, and emotive expressions) to replace the analogous methods. They survey the different modalities of interacting with CAD data, discussing the unimodal category (eye-tracking, gesture, speech, and brain-computer interfaces) first and then looking at the interaction between each. Their survey results indicated the "Gesture + Speech" category has the highest body of research (59.3%). Their survey also included the available devices for natural HCI that can be used in CAD programs. Overall, the paper is a comprehensive literature review of the current state of the art methodologies present in the field of HCI in CAD. One aspect it does not address is the software interface's HCI considerations in CAD software.

Bahr et. al. [22] investigate the advancements in CAD software for problem solving, as well as identify research gaps to address features that affect cognitive psychology, interaction design, and design engineering.

The authors reviewed two major CAD packages (SolidWorks by Dassault Systèmes and Creo by PTC) and attempted to relate the cognitive psychology domain to CAD. One of the advantages they point out is that automation of solutions in these software saves time, however it can end up detaching the engineer from creative reasoning. [22] The reinforcement of functional fixedness needs to be alleviated in such tools. They conclude by stating four key design principles to integrate design needs into a CAD system, to capture the reasoning of a design [1]:

a. Tracking, capturing, and storing design reasoning with the engineering aspects of the part
b. Tracking all design solution possibilities, provide comparisons and advice, and learn new solutions with assistance (from designer)
c. Standard parts' functionalities should be included
d. Retrieve past features or parts from previous sessions and capture the intent of design.

This displays that researchers are working on making CAD design tools more intuitive and utilize multiple modalities to help the user to the maximum extent possible. This allows the creator of a CAD part to capture design intent and thought processes, when building the part, the evaluator can then use to evaluate the part using a similar intuitive interface.

One of the major gaps that is not addressed by the existing body of research is the software's interface and how it affects the cognitive load on users. The applications support design and development but not the design intent built into parts. [22] Another important aspect of understanding the HCI implications in CAD is to understand how the various processes involved in CAD can be utilized to tailor the HCI design guidelines for the application interface.

2.2 Assessing and Evaluating CAD Models

The techniques to assess CAD models are heavily reliant on visuomotor skills, where the cognitive load gets enhanced the more complex the models are. [20] The design intentions frequently face disruptions due to the switchbacks between the different features. [21] The same is applicable for evaluation of CAD models as well. As the complexity increases, the depth at which the model must be evaluated to preserve design intent and functionality also increases. This indicates the need to ensure the process of understanding CAD information is intuitive.

The automatic assessment of CAD geometry is a novel technique that is currently used to evaluate the geometry and changeability of a model. [24] Kirstukas [24] in their study compared student files to a "gold standard" model and found strong correlation between scores achieved after traditional grading and automatic grading. The author stated that the algorithm was able to differentiate between average and excellent models, and the time saved was in the hundreds. Thus, this could indicate that with the right evaluation techniques, a lot of time could be saved during evaluation and the design intent could also be preserved. The drawback of this tool is the degree of resolution (decimal point accuracy) could lead to false positives, and multiple of the same feature types might get stacked together, which again cannot show the true picture.

Johnson et. al. [25] in their paper investigate and evaluate the metrics associated with CAD models. Their argument is that there exists no standardized and objective complexity metrics, for part or process evaluation. [26] To accomplish part evaluation, the most common processes are usually ad-hoc, but also repetitive. This is due to the nature of CAD parts, where unless the design intent is specifically requested of, there are multiple paths to the correct end goal. Johnson et. al. [25] used 10 test components for testing complexity and 47 experimental components to test the relationship between complexity, model attributes, and modelling time. Their results demonstrated volume ratio correlated with model complexity. Modelling time also increased with the increase in features, thus an increase in time to evaluate.

This literature body helps shed light on the repetitive and ad-hoc nature of evaluating CAD models. There is a need to understand the core element of a CAD part that gets evaluated and redesign the process around it. In that way, there is an increase in time saved, regardless of geometry complexity.

2.3 Utilizing Industry Technology as Educators

The role of industry technology is a crucial one, when implementing engineering curricula that facilitates "Industry 4.0". The implementation of industry technology can be helpful for learners and educators alike. Arlett et. al. [11] in their research examined the core principles behind "experience-led engineering degree". Their study looked at six different universities' engineering programs, and amidst all the differences there are common elements to integrate problem-based learning (PBL) in the classroom. They state that academic staff with industry experience are crucial to the learning of vital skills by the student. However, there has been a decrease in such faculty with relevant experience in institutes. The lack of adept evaluators can cause improper evaluation of the current student output, thus failing to achieve the goal of "experience-led engineering degree". Thus, a standardized evaluation tool, following Industry 4.0 guidelines, could provide training to evaluators to get experienced with an industry environment, in turn then providing students with professional feedback.

Andersson and Andersson [12] conducted a role play experiment, where students were engaged in industry involvement. Role play experiments have been effective in getting students to learn the underlying context of the engineering ecosystem, the stakeholders involved, and the dynamic complexities involved in solving engineering problems. Van Mets [13] describes in their book the benefits of learning in a realistic environments and its ability to encourage participation in the learning process. The usage of engineering processes, terminologies, tools, and context enables such role play in learners. Andersson and Andersson [12] found that the students were able to express an innate understanding of their roles in an engineering environment, when exposed to role-play scenarios, with the participation of industry figures. Thus, academic evaluators (Graduate Teaching Assistants specifically) could benefit from role-playing as if they were part of a real organization, evaluating their employees' (their students') outputs, which leads to a professional and standardized feedback and scoring environment.

Industry technology also encapsulates working in interdisciplinary teams to accomplish objectives. Interdisciplinary engineering education is geared towards teaching students with varied expertise, to solve goals, by working within the same context. [14,15] Most interdisciplinary education is focused on collaborative teamwork. Beemt et. al. [5] found that one of the major drawbacks when developing a structured education system that relies on interdisciplinary education, is the lack of industry expertise in educators. The course evaluators (especially Graduate Teaching Assistants) might belong to a variety of engineering backgrounds, thus each having their own evaluation mindset and design workflow for a CAD part. The development of a standardized workflow and tool enables multi-disciplinary evaluation teams to assist with student work evaluations, in a streamlined and standardized manner.

3 SYSTEM DESIGN & DESIGN PROCESS

The proposed solution is a custom evaluation tool, that is developed using HTML, CSS, Javascript for the frontend, and Django for the backend solutions. The application is meant to be deployed as a website for satisfying evaluation needs, of a CAD evaluator.

3.1 User Persona

The first step is to develop a User Experience (UX) Persona Model for the ideal user of this system. The persona has been developed after informal interviews, observations, and expert testimonials from current evaluators in a course with learning objectives in line with Industry 4.0. Table 1 provides an example of an individual from the the targeted user base. This User Persona was developed using Contextual Inquiry [28] of Graduate Teaching Assistants of a CAD course.

Table 1: UX Persona for John, target audience for the system

Property	Related Data
Age	24-28
Employment Status	Graduate Teaching Assistant
Education	Bachelors and Masters in Engineering
Technical Ability	Proficient in CAD; computer operation; mathematical analysis; spatial analysis; reasoning ability
Behaviors	Follows instructions closely; is a team player; stays up to date with course material (for his TA duties); helps onboarding of other TAs
Pain Points	Spends too much time grading; spends a lot of effort to set up multiple documents to remember grading rubric; sometimes is confused about score so does a random deduction; grades inconsistently when deadline is close; is shy to ask for help frequently
Goals	Wants to grade more consistently (even for the entire team); wants to reduce time spent on grading;
Needs	a single interface to access grading, rubrics, and instruction notes; a clean way to switch between assignments, easy answer key reference; standardized feedback mechanism; standardized language for grading

3.2 Application Design

The User Persona model dictates the core functionality of the web-based tool. This is helpful to determine the primary design goals to address the research questions. The design goals have been categorized. into two problem spaces to address – TIME and PROCESS. The primary design goals can be listed as follows, with the priority, importance, and development order’s cumulative score marked as either A (highest), B (medium), or C (lowest):

a. TIME and PROCESS: Responsive web page (B) following a “star” pattern for information architecture (A) where the evaluator can open different assignments to evaluate from a single screen.
b. PROCESS: The Landing page has a password protected login screen. (B) Once logged in, the next page has a “Week-at-a-Glance” feature, allowing evaluators to quickly sift through a semester’s workload by filtering it using a dropdown. This page also has administrative login controls, that

allows the administrator to create tables (for models) to be used within the application, using a graphical user interface of Django's built in CRUD (Create-Read-Update-Delete) functionality.

c. TIME: The main "Contents" page has an accordion style dropdown with all the semester's assignments. This page functions as a control hub, and when a week is clicked on, it expands to display all the assignments from that week. Once the assignments are revealed, clicking on them opens a sub-pane on the right side, that displays details of the assignment and has a button to launch the "Evaluation" screen for that assignment.

d. TIME and PROCESS: The "Evaluation Screen" has three major elements:
    i. Vertical navigation bar (collapsable) that links to all assignments (A), resources, and "Sign-Out" button
    ii. On the left side panel of the screen is an image of the CAD model, and a button that opens up a modal. The modal contains an interactable image that allows for switching between the original and morphed model. The modal also contains important model metadata.
    iii. On the right-side panel of the screen is an interactable table (with dropdowns) that allows the scores to be manipulated in real-time. The table is automatically populated based on tables (models) created in the administrator view. There are four options in the dropdown to change scores, based on the most common needs by an evaluator:
        a. TIME: Half-score Deduct – Reduces the score of the specific criteria by 50% (A)
        b. TIME: Full-score Deduct – Reduces the score of the specific criteria by 100% (A)
        c. TIME: Default score – Keeps the score at 100% as a default option (A)
        d. TIME: Custom score – Changes the score column to an editable box, to directly manipulate values
    iv. Finally, clicking on a "Generate Feedback" button compiles all the altered scores, and generates feedback (based on the rubric type) that can be copied.

The TIME problem space is evaluated by measuring total time taken and section time taken, for short evaluation scenarios of a part. The PROCESS problem space is evaluated by measuring self-reported count of "back-and-forth" moments and using Likert Scale's self-reported scores. Appendix B illustrates some of the screenshots of the application.

4 EVALUATION METHODOLOGIES

4.1 Eligible Participants

The eligibility criteria of participants are determined based on the ideal User Persona stated in Section 3. The participant must possess knowledge of working with CAD software (preferred software are Siemens NX and Dassault SolidWorks). The participants are recruited in two phases. Phase I participants are heavily involved in the co-design of the application. They must be experts of CAD as well as expert evaluators, ideally Graduate Teaching Assistants in a CAD course, with at least 1 year (or two semesters) held in that role. Phase II participants are used to validate the tool and have less strict requirements (evaluator with at least one semester or four months of CAD experience and current CAD students).

4.2 Apparatus

The Testing Methodologies that were used to evaluate the application are:

a. Think Aloud Protocol [9] – This is the primary tool of feedback after each development iteration, from Phase II participants.

b. Heuristic Evaluation (10 Standards by Jakob Nielsen) [10] – This is the primary tool of design and iteration, with Phase I participants (co-design with experts), for the web interface of the system. The evaluation is administered via a Qualtrics Survey and Microsoft Excel.

c. Data Analytics on Performance Metrics (Overall Time, Evaluation Time, and Error Rate) – This is recorded during Phase II. The overall time determines the average time-to-completion of evaluation of a standard CAD part. The Evaluation time directly addresses the research question (RQ1).

4.3 Procedure

a. Phase I – The application iterations are developed in an agile format, with a one-week sprint. At the end of each sprint, the developer does a sit-down interview with one participant (expert), while the participant does a simple evaluation of a CAD part. The evaluation has two components: first, it directly utilizes the newest developed feature for that sprint directly, and then the evaluation from the beginning is redone (following the set process). Think-aloud protocol is used to find the major highlights and low points for the next iteration.

b. Phase II – Towards the end of the development cycle (around the end of eight sprints), the application is presented to the participants (novice). They are provided with a text rubric, and then asked to grade a mock CAD part submission. The evaluation begins from the initial stage of logging into the application and then ends when they finalize a numerical score with textual feedback, for the provided submission. Total time taken is noted, and then a post-experiment Likert Scale (5-point) questionnaire is asked to record quantitative data (Appendix A).

The participants grade one model (designed in CAD software Siemens NX). The part comprises of eight features (three extrudes with one sketch each). The participants start with the part loaded in, as well as the requirements document open. Once the investigator gives the “start” signal the evaluators they must evaluate the part in the form of three tasks:

a. Check the mass and material, and set the feature 1 score with feedback

b. Morph two dimensions in the base sketch and check the constraint status of the sketch. Then set the feature 2 score with feedback.

c. Check one more additional rubric item, grade it, and set the feedback.

d. Once evaluation is complete, copy the feedback and score to a Learning Management System (Brightspace).

Each section has its time (T) noted, as well as the number of times the evaluator went back and forth (B) the requirements document and the grading tool (manual text tracker or the website application). The total time taken and total number of “back-and-forths” are also recorded, to derive insights for the research question. Once the evaluation session is complete the participants answer a post-experiment Likert Scale (5-point) questionnaire (Appendix A).

5 RESULTS

A small user study was conducted to evaluate the feasibility and usability of the web application. The results were collected for experts (Phase I) and novice (Phase II) evaluators. The results are listed in Table 2.

Table 2: Results of user study (three expert users and three novice users)

Average / Roles	Manual Total Time (T_m seconds)	Website Total Time (T_w seconds)	Manual "back-and-forths" (B_m)	Website "back-and-forths" (B_w)
Experts (Phase I)	121	83.34 *(↓ 31.13%)*	4.67	3
Novice (Phase II)	169	125.67 *(↓ 22.29%)*	8.67	6.67

The results display a reduction in the total time needed for one part to be evaluated. It can also be seen that the number of "back-and-forths" between the evaluation screen and requirements document is reduced significantly, saving time as well. This indicates the potential for the tool to be used for evaluation in large scale CAD courses. Both phase participants did share the sentiment that, with more time dedicated to development, this would be vital tool that they would use for their daily evaluation operations.

6 DISCUSSION

The results do indicate that there is a potential to reduce the time involved in evaluation of CAD models, when utilizing a combined interface, that has been co-designed with CAD experts. From the think aloud sessions, as well as the survey responses, there was a clear indication that the web application was aligned with their current training and expectations for the most part. The scores for "Saved me time" and "Reduces need to remember information" scored very high, indicating that the web application can reduce the stress induced due to having to switch between multiple documents and rubric calculations, during grading.

What went well in the application was highlighted from the positives cited by the evaluators hinted towards the beginning of a solid foundation that can be improved over time. The evaluators were enthusiastic about having an interface that is closely related to the course material, and yet is flexible enough to accommodate for future expansions. Another benefit of the system is that it enabled the evaluators to slowly start trusting the system and reducing the number of cross-checking needed to ensure they gave the feedback correctly. For researchers and developers wanting to develop a similar system, one piece of advice would be to build around a core functionality that the evaluators will be doing for most of the time. Grading is the primary activity, in this case, and thus the web application was built around the real-time score and feedback generation table. This allowed for more time to be spent on the critical components, and then built outwards, based on the team's needs. The evaluators graded the "overall usability" of the application as an average score of 3.3 (out of 5).

The questionnaire also revealed some of the limitations and desired features of the application. The lowest scores were achieved in the category of "warns about potential errors", "provides shortcuts to aid me", and "explains error in an understandable manner". This shows that the application lacks critical onboarding features as well as guiding elements (such as tooltips, help icons, focus elements etc.) which could be a failure point for potential users. This could also deter the users from fully adopting and trusting the system and must be addressed as soon as possible. Finally, another limitation is the application has not implemented the evaluation mechanics for assemblies and large projects – which is another major aspect of CAD education. These will be addressed in future iterations.

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APPENDIX A

The questions used in the questionnaire are provided below. The questionnaire has been created using Qualtrics and data was analyzed using Microsoft Excel.

5-point Likert Scale from 1 (Strongly Disagree) to 5 (Strongly Agree)

1. The system was easy to use.
2. The system saved me (the user) time when evaluating.
3. The system is intuitive to use.
4. The system had all the expected components that I (the user) require for evaluation.

5-point Likert Scale (Jakob's Heuristics)

1. The system provides proper feedback to my (the user's) actions. (Visibility)
2. The language used by the system would typically be found in a CAD environment. (Matching System and Real World)
3. The system allows me (the user) to fix an action that was done as a mistake and take a step back. (User Control and Freedom)
4. The system is consistent in meaning with the available actions and industry standards. (Consistency and Standards)
5. The system warns me (the user) about potential errors that might occur. (Error Prevention)
6. The system reduces my need to remember information and action i.e. it clearly makes elements, actions, and options visible and readable. (Recognition Rather than Recall)
7. The system provides shortcuts to aid me (the user). (Flexibility and Efficiency of Use)
8. The system has does not overload me (the user) with information that is not relevant. (Aesthetic and Minimalist Design)
9. The system explains errors in an understandable manner. (Help Users Recognize, Diagnose, Recover from Errors)
10. The system provides access to help and documentation when needed. (Help and Documentation)

5-point Likert Scale (Overall Score)
Overall Score for Usability as a viable tool for evaluating CAD models.

APPENDIX B

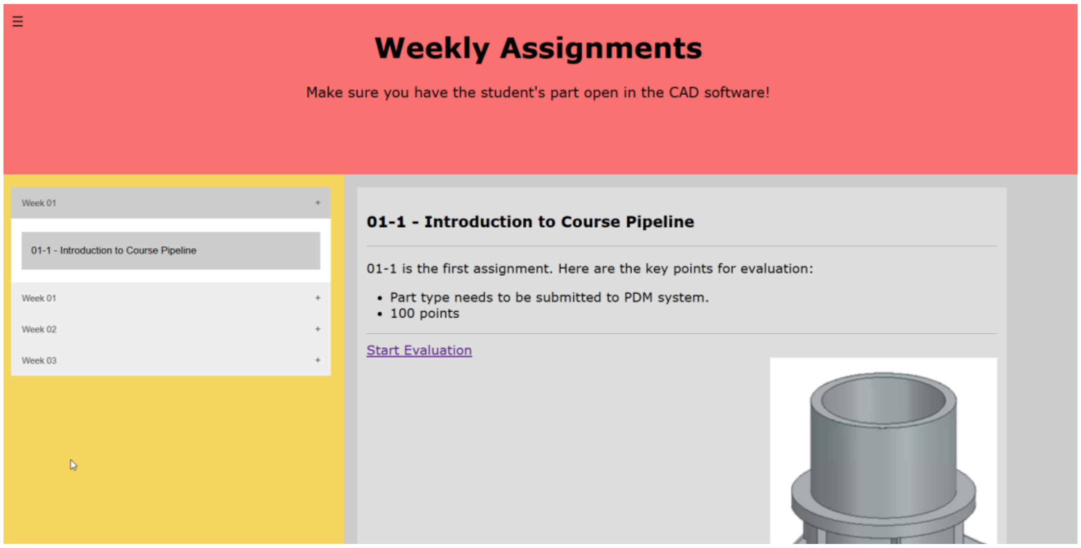

Figure 1: The index page that contains links to all assignments of the semester and clickable links to the evaluation screen.

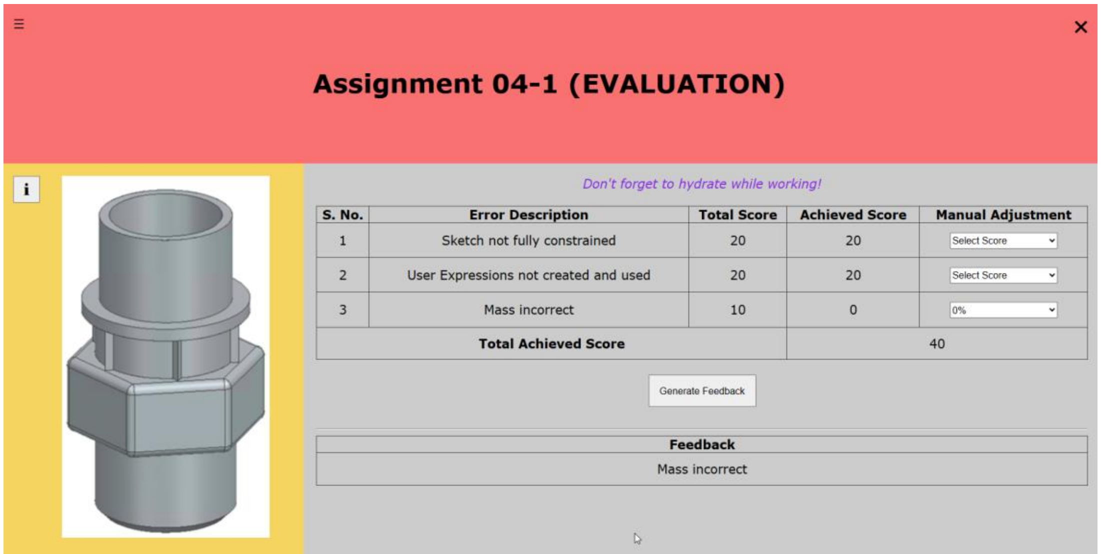

Figure 2: The initial evaluation screen that contains the modal button, the original part image, the real-time rubric table, feedback zone, and real time score calculations.

Don't forget to hydrate while working!

S. No.	Error Description	Total Score	Achieved Score	Manual Adjustment
1	Sketch not fully constrained	20	10	50%
2	User Expressions not created and used	20	20	100%
3	Mass incorrect	10	0	0%
4	Wrong orientation	20	9	Custom
Total Achieved Score			39	

Generate Feedback

Figure 3: The rubric table (generated from models) with real time score and feedback updates.

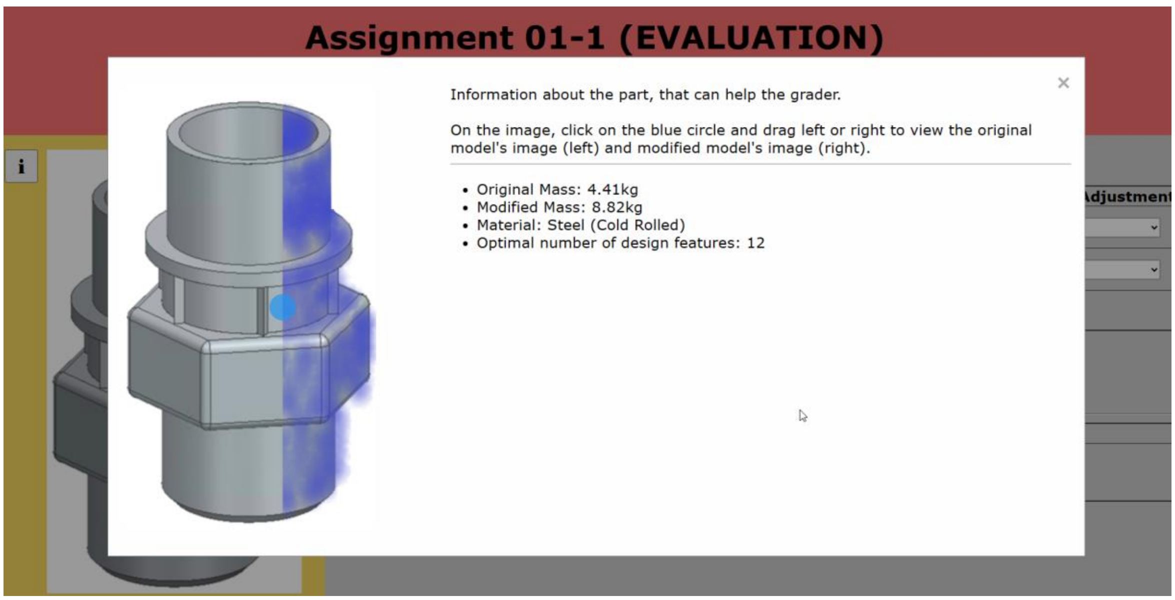

Figure 4: Clicking on the “i” button on the image opens a model, allowing evaluators to dive deeper into the metadata. Also contains an interactable image (left side shows original model and right side shows what it should look like after morphing)